Algebra 1: Unit 8, Lesson 7

1. Sam is rewriting the equation, $y = -3x^2 - 6x - 8$, in vertex form by factoring the leading coefficient from the first two terms. What is the new expression after this step?

2. An expression is shown.

   $x^2 - 12x + c$

   Which is the value of c needed to make the expression a perfect square trinomial?

3. Given the steps:

   $y = 5x^2 + 30x + 7$
   $y = 5(x^2 + 6x) + 7$
   $y = 5(x^2 + 6x + 9) + 7 - n$
   $y = 5(x + 3)^2 + 7 - n$
   $y = 5(x + 3)^2 - 38$

   What is n and how is it found?